



McGill

Design Team Internship Timeline

Anne-Sophie Nadeau

Luca Abruzzese

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2026 Design Team Internship Timeline

Week 1: May 11 – May 17

- Analyze Multimatic damper testing data and generate initial findings report
- Develop a preliminary testing and validation plan for the 7 DOF vehicle model
- Create and finalize the GitHub repository structure, workflow, and documentation standards
- Begin research into quasi steady-state lap simulation methodologies and existing approaches used in motorsport

Week 2: May 18 – May 24

- Continue research into quasi steady-state lap simulation theory and implementation methods
- Develop the initial UML architecture diagram defining class structure, object interactions, and data flow
- Begin development of the half-car ride model in Simulink

Week 3: May 25 – May 31

- Continue development of the half-car Simulink model
- Begin implementation of the 7 DOF full vehicle model
- Implement heave, pitch, and roll dynamics into the full vehicle model

Week 4: June 1 – June 7

- Finalize initial implementation of the 7 DOF vehicle model
- Verify model stability and response behavior under standard vehicle 4 post inputs
- Prepare validation procedures for model correlation

Week 5: June 8 – June 14

- Validate half-car and 7 DOF models using analytical checks, expected dynamic trends, and available test data from Multimatic
- Compare simulated responses against theoretical vehicle dynamics behavior
- Refine damping, stiffness, and tire parameters for improved correlation
- Document assumptions and model limitations

Week 6: June 15 – June 21

- Michigan Formula SAE Competition
- Gather vehicle observations, setup data, and driver feedback for future model improvements
- Analyze real world vehicle behavior to guide future simulation refinement

Week 7: June 22 – June 28

- Finalize UML architecture and overall software structure
- Begin coding the quasi steady-state lap simulation framework
- Implement vehicle state handling and track segmentation logic

Week 8: June 29 – July 5

- Continue development of the quasi steady-state lap simulation
- Implement longitudinal acceleration and braking performance calculations
- Integrate tire force and powertrain limitations into the simulation
- Begin development of cornering performance calculations

Week 9: July 6 – July 12

- Continue coding and integration of lap simulation modules
- Implement lateral acceleration limits and combined tire loading behavior
- Develop data logging and visualization tools
- Begin debugging and verification of lap simulation outputs

Week 10: July 13 – July 19

- Continue refinement of quasi steady-state lap simulation
- Improve numerical stability and computational efficiency
- Integrate aerodynamic effects and load sensitivity behavior
- Begin sensitivity studies on suspension and vehicle parameters

Week 11: July 20 – July 26

- Finalize primary lap simulation functionality
- Perform parameter sweeps and correlation studies
- Compare simulation trends against expected vehicle dynamics behavior
- Prepare validation procedures and test cases

Week 12: July 27 – August 2

- Validate quasi steady-state lap simulation using analytical calculations and available vehicle data
- Compare predicted lap behavior against expected vehicle performance trends
- Refine assumptions and vehicle parameter models
- Document validation methodology and results

Week 13: August 3 – August 9

- Continue lap simulation validation and refinement
- Perform sensitivity analysis on suspension, tire, and aerodynamic parameters
- Improve usability and modularity of the simulation framework
- Finalize model outputs and reporting tools

Week 14: August 10 – August 16

- Begin final project documentation
- Document methodologies, assumptions, model structures, and validation procedures
- Organize repository structure and supporting technical material
- Prepare figures, plots, and simulation results summaries

Week 15: August 17 – August 23

- Continue final documentation and technical reporting
- Finalize UML diagrams and software architecture documentation
- Create user guides and model usage instructions
- Review overall project completeness and technical accuracy

Week 16: August 24 – August 30

- Complete final documentation and repository cleanup
- Final review of models, simulation tools, and validation results
- Address remaining issues, bugs, or missing documentation
- Buffer week for additional refinements, testing, or unforeseen delays